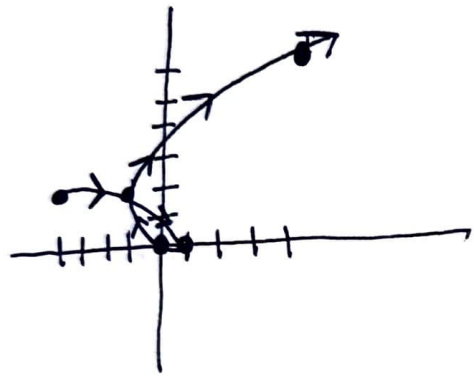


Unit 6 KEY

1. Given $\begin{cases} x(t) = t^3 - 2t \\ y(t) = t^2 + t \end{cases}$

a) Make a table for $t = \{-2, -1, 0, 1, 2\}$

b) Graph the equation for $-2 \leq t \leq 2$



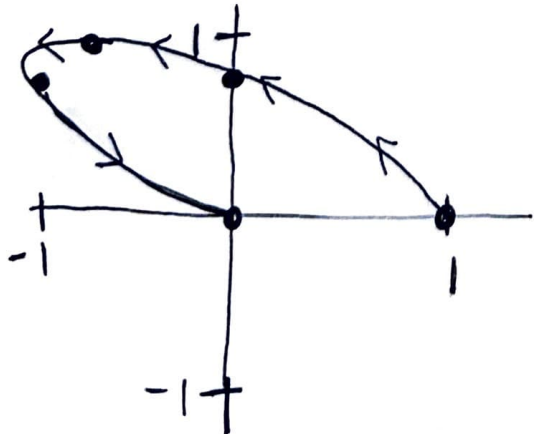
t	$x(t)$	$y(t)$
-2	-4	2
-1	-1	0
0	0	0
1	-1	2
2	4	6

2. Given $\begin{cases} x(t) = \cos(3t) \\ y(t) = \sin(2t) \end{cases}$

a) Make a table for $t = \{0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}\}$

b) Graph the equation for $0 \leq t \leq \frac{\pi}{2}$

t	$x(t)$	$y(t)$
0	1	0
$\pi/6$	0	$\sqrt{3}/2 \sim 0.87$
$\pi/4$	$-\frac{\sqrt{2}}{2} \sim -0.7$	1
$\pi/3$	-1	$\sqrt{3}/2 \sim 0.87$
$\pi/2$	0	0



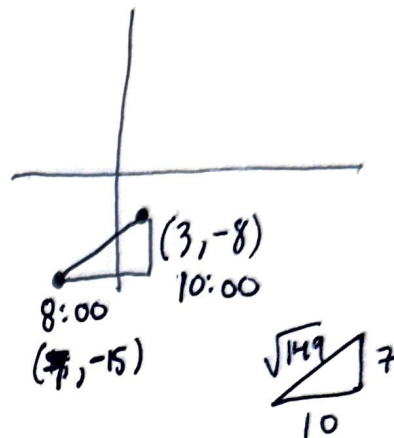
c) Go check out Desmos!

It's COOL!!

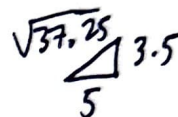
3. At 8:00 am, boat is 15 mi S, 7 mi W
 At 10:00 am, boat is 8 mi S, 3 mi E

a) What is Dist. traveled in one hour?

Boat travels $\sqrt{37.25} \sim 6.1$ mi
 (or $\frac{1}{2}\sqrt{149}$)



b) At 12:00 pm, travel 10 mi E and 7 mi N
 of position at 10:00 am



Boat is 13 mi E and 1 mi S of
 the port at noon (Location is $(13, -1)$)

c) Boat Starts 7 mi West, travels 5 mph East (x-component)
 Boat Starts 15 mi South, travels 3.5 mph North (y-component)

$$\begin{cases} x(t) = -7 + 5t \\ y(t) = -15 + 3.5t \end{cases}$$

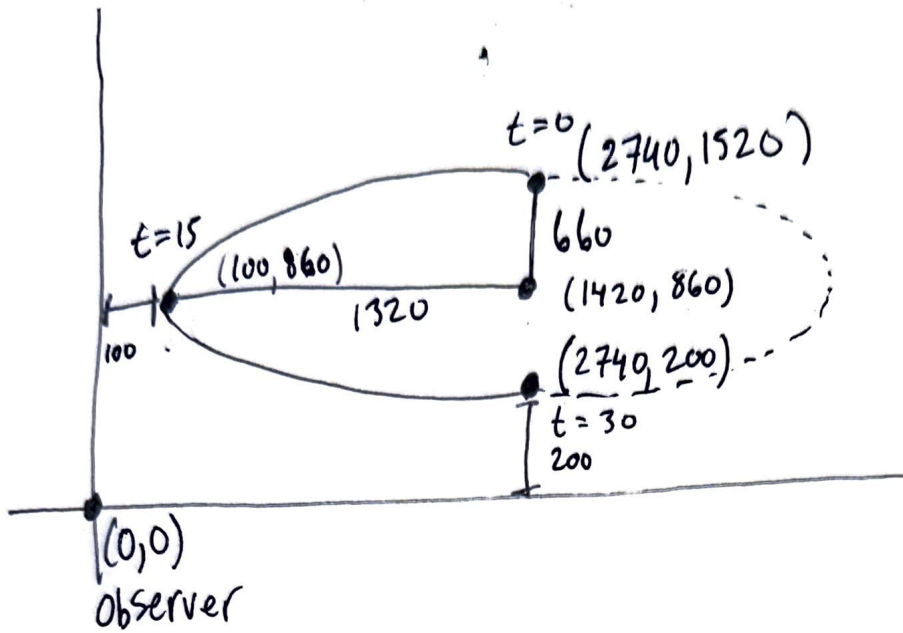
d) 4:00pm is 8 hours after 8:00 am

$$x(8) = -7 + 5(8) = -7 + 40 = 33$$

$$y(8) = -15 + 3.5(8) = -15 + 28 = 13$$

At 4:00pm, the boat
 is 33 mi East and
 13 mi North of the port

4.



$$a) \quad \frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \quad (h,k) = (1420, 860)$$

$$a = 1320 \quad b = 660$$

$$\boxed{\frac{(x-1420)^2}{1320^2} + \frac{(y-860)^2}{660^2} = 1}$$

b) $t=0$ starts at $(x,y) = (2740, 1520)$
Car travels counter-clockwise

Period is 60 seconds
So $B = \frac{2\pi}{60} = \frac{\pi}{30}$

X-value goes from middle $\rightarrow$ min $\rightarrow$ middle $\rightarrow$ max
So use $-\sin$ for $x(t)$

Y-value goes from max $\rightarrow$ middle $\rightarrow$ min $\rightarrow$ middle
So use $+\cos$ for $y(t)$

$$\frac{x-1420}{1320} = -\sin(t)$$

$$\frac{y-860}{660} = \cos(t)$$

$$\boxed{\begin{cases} x(t) = 1420 - 1320 \sin\left(\frac{\pi}{30}t\right) \\ y(t) = 860 + 660 \cos\left(\frac{\pi}{30}t\right) \end{cases}}$$

5. Rewrite the rectangular equation (given below) as a system of parametric equations, $x(t)$ and $y(t)$.

$$\frac{(x+3)^2}{100} + \frac{(y-1)^2}{16} = 1$$

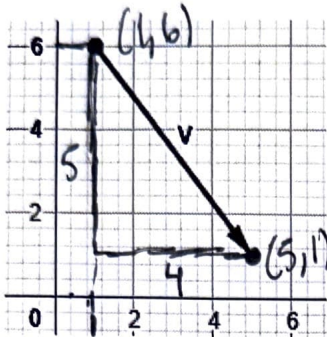
$$\sin^2 t + \cos^2 t = 1$$

$$\text{Let } \frac{x+3}{10} = \sin t \Rightarrow$$

$$\text{Let } \frac{y-1}{4} = \cos t \Rightarrow$$

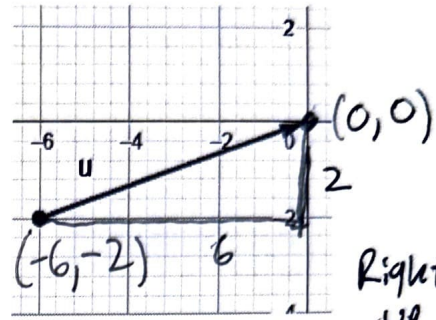
$$\begin{cases} x(t) = 10 \sin t - 3 \\ y(t) = 4 \cos t + 1 \end{cases}$$

6. Answer the following given the image of vectors $\vec{v}$ and $\vec{u}$.



Right 4
Down 5

It is a little hard to see, but vector $\vec{v}$ ends at (5, 1).



Right 6
Up 2

a) Find the component form of $\vec{u}$ and $\vec{v}$.

$$\vec{v} = \langle 4, -5 \rangle \text{ and } \vec{u} = \langle 6, 2 \rangle$$

b) Find the direction of $\vec{u}$ and $\vec{v}$ (center each vector at the origin and find $\theta_{\vec{u}}$ and $\theta_{\vec{v}}$).

$$\theta_{\vec{v}} = 308.66^\circ$$

$$\theta_{\vec{u}} = 18.43^\circ$$

c) Write the component form of $4\vec{u} - 3\vec{v}$.

$$4\vec{u} - 3\vec{v} = \langle 12, 23 \rangle$$

7. Given each vector, find its magnitude and direction.

a) $\vec{u} = \langle 10\sqrt{3}, 10 \rangle$

$$\|\vec{u}\| = 20, \theta_{\vec{u}} = \pi/6$$

b) $\vec{v} = \langle 2\sqrt{2}, -2\sqrt{6} \rangle$

$$\|\vec{v}\| = 4\sqrt{2}, \theta_{\vec{v}} = 5\pi/3$$

c) $\vec{w} = \langle -4, 4 \rangle$

$$\|\vec{w}\| = 4\sqrt{2}, \theta_{\vec{w}} = 3\pi/4$$

8. Graph each coordinate (r, θ) on the polar coordinate system given.

a) $(3, \frac{\pi}{4})$

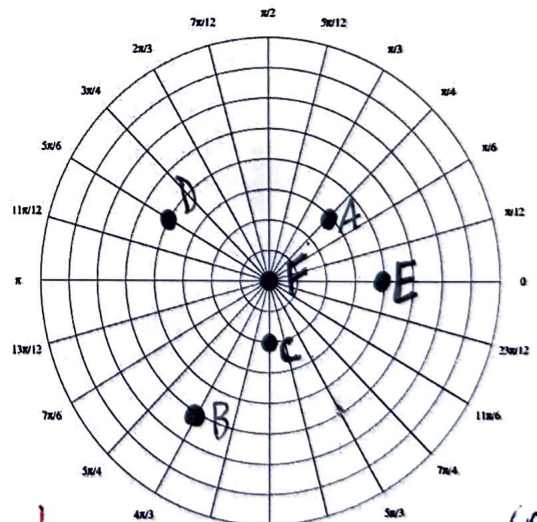
d) $(-4, -\frac{\pi}{6})$

b) $(5, -\frac{2\pi}{3})$

e) $(4, 0)$

c) $(-2, \frac{\pi}{2})$

f) $(0, 4)$



9. Convert between rectangular coordinates and polar coordinates.

	(x, y)	(r, θ)
1)	$(-2\sqrt{2}, 2\sqrt{2})$	$(4, \frac{3\pi}{4})$
2)	$(-\sqrt{3}, -1)$	$(2, \frac{7\pi}{6})$
3)	$(3, 0)$	$(3, \pi)$
4)	$(0, 5)$	$(5, \pi/2)$

4) $(0, 5) = (x, y)$
Same as
 $(r, \theta) = (5, \pi/2)$

$$1) x = r \cos \theta$$

$$y = r \sin \theta$$

$$\Rightarrow x = 4 \cos \frac{3\pi}{4} = 4 \left(-\frac{\sqrt{2}}{2} \right) = -2\sqrt{2}$$

$$y = 4 \sin \frac{3\pi}{4} = 4 \left(\frac{\sqrt{2}}{2} \right) = 2\sqrt{2}$$

$$2) r^2 = x^2 + y^2$$

$$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6} \text{ or } \frac{7\pi}{6}$$

$$3) (r, \theta) = (-3, \pi)$$

Same as
 $(3, 0) = (x, y)$

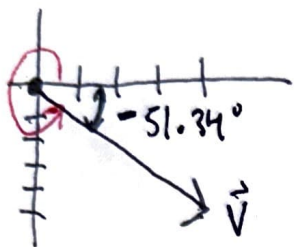
6. b) Given $\vec{v} = \langle 4, -5 \rangle$ and $\vec{u} = \langle 6, 2 \rangle$

$$\tan \theta_v = \frac{y}{x} = \frac{-5}{4}$$

$$\theta_v = \tan^{-1}(-5/4) \approx -51.34^\circ$$

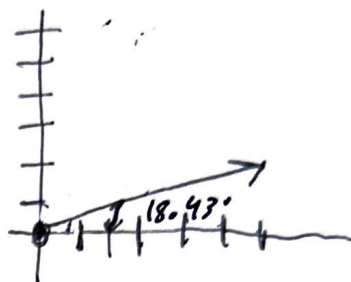
$$\tan \theta_u = \frac{2}{6} = \frac{1}{3}$$

$$\theta_u = \tan^{-1}(1/3) \approx 18.43^\circ$$



$$-51.34^\circ + 360^\circ = 308.66^\circ \text{ (co-terminal)}$$

$$\therefore \theta_v \approx 308.66^\circ$$



$$\therefore \theta_u \approx 18.43^\circ$$

7. Find magnitude and direction

a) $\vec{u} = \langle 10\sqrt{3}, 10 \rangle$

$$\|\vec{u}\| = \sqrt{(10\sqrt{3})^2 + 10^2}$$

$$= \sqrt{3000 + 100}$$

$$= \sqrt{4000} = 20$$

$$\therefore \|\vec{u}\| = 20$$

$$\tan \theta_u = \frac{10}{10\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\theta_u = \frac{\pi}{6} \text{ or } \frac{7\pi}{6} \text{ but } \vec{u} \in \text{Q I}$$

$$\therefore \theta_u = \pi/6$$

b) $\vec{v} = \langle 2\sqrt{2}, -2\sqrt{6} \rangle$

$$\|\vec{v}\| = \sqrt{(2\sqrt{2})^2 + (-2\sqrt{6})^2}$$

$$= \sqrt{8 + 24} = \sqrt{32}$$

$$\therefore \|\vec{v}\| = 4\sqrt{2}$$

$$\tan \theta_v = \frac{-2\sqrt{6}}{2\sqrt{2}} = -\sqrt{3}$$

$$\theta_v = \frac{2\pi}{3} \text{ or } \frac{5\pi}{3}$$

$$\text{but } \vec{v} \in \text{Q IV}$$

$$\therefore \theta_v = \frac{5\pi}{3}$$

$\vec{w} = \langle -4, 4 \rangle$

$$\|\vec{w}\| = \sqrt{(-4)^2 + 4^2}$$

$$= \sqrt{16 + 16} = \sqrt{32}$$

$$\therefore \|\vec{w}\| = 4\sqrt{2}$$

$$\tan \theta_w = \frac{4}{-4} = -1$$

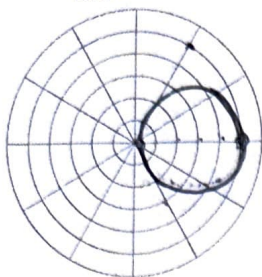
$$\theta_w = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}$$

$$\vec{w} \in \text{Q II}$$

$$\therefore \theta_w = \frac{3\pi}{4}$$

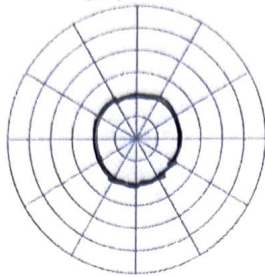
10. Sketch each polar curve given the equation.

circle



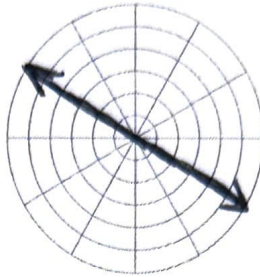
a) $r = 5 \cos \theta$

circle



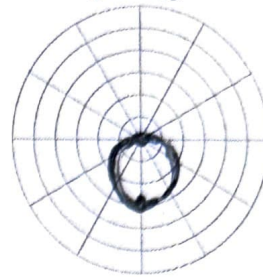
b) $r = 2$

Line

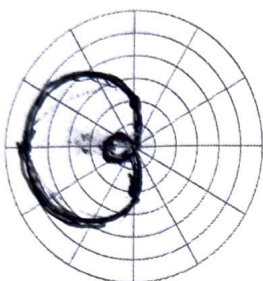


c) $\theta = 5\pi/6$

Circle

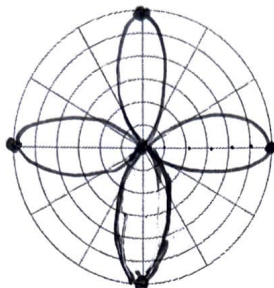


d) $r = -3 \sin \theta$



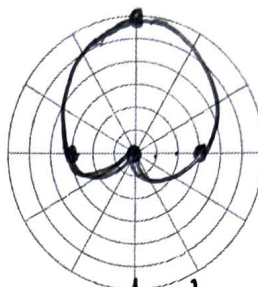
e) $r = 2 - 3 \cos \theta$

Limaçon
Inner Loop



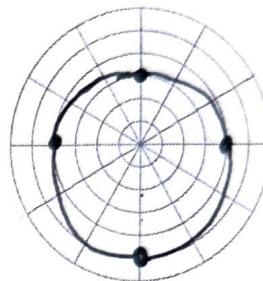
f) $r = 6 \cos 2\theta$

Rose Curve



g) $r = 4 + 4 \sin \theta$

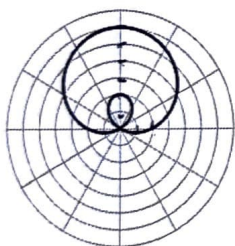
Cardioid
Change to
 $r = 3 + 3 \sin \theta$
so it will graph nice



h) $r = 4 - \sin \theta$

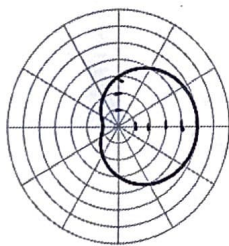
Convex
Limaçon

11. Write the polar equation given the graph of each curve.



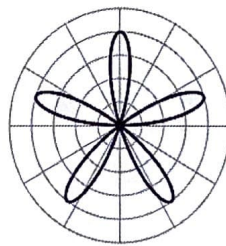
a) $r = 2 + 4 \sin \theta$

Limaçon
Inner Loop



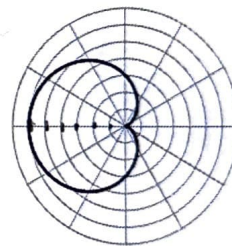
b) $r = 3 + 2 \cos \theta$

Dimpled
Limaçon



c) $r = 4 \sin(5\theta)$

Rose Curve



d) $r = 3 - 3 \cos \theta$

Cardioid